CS 5330 HW6

1. What is the difference between a moment and a central moment of a region?

   The default moment is relative to the standard coordinate system. The central moment is relative to the body. Default moments are only useful because they can be used to calculate the mean x/y coordinate of an object, which are in turn used to compute the central moments.

2. Why is it useful to compute the axis of least central moment for a region?

   with the least central moment vector known, you can find the minimum and maximum projections, which let you find the most optimal bounding box around the object. This in turn can be used for a variety of purposes, including bounding-box-based matching (comparing other objects' based on metrics that utilize the bounding box).

3. What is the difference between nearest neighbor and k-nearest neighbor matching?

   In a matching / classification problem, nearest neighbor uses only the top match / class (k=1), while k-nearest neighbor uses multiple matches. K-nearest gives a more stable prediction since it gives more matches, or "votes".

4. When is it important to use scaled Euclidean distance instead of standard Euclidean distance?

   If different features you are comparing have different scales (as is often the case), it is crucial to combine them so that one feature does not dominate the distance metric computation.

5. Reflection Models Speed Round:

   a. True or False: body reflection depends on the viewer's location.

      False

   b. True or False: surface reflection depends on the viewer's location.

      True

c. True or False: the intensity of Lambertian reflection is proportional to the dot product of the surface normal and the viewing direction.

   False, it is the Light direction (L) dotted with the surface normal (N)

d. True or False: metals do not exhibit body reflection.

   True

e. True or False: body reflection is caused by photons penetrating the surface and interacting with pigment particles inside the material.

   True

f. True or False: velvet looks different because of quantum interactions in the material.

   False

g. True or False: velvet looks different because the surface normals of the fibers are perpendicular to the overall surface normal.

   True

6. What makes the bi-illuminant dichromatic reflection model different from the standard dichromatic reflection model.

   the bi-illuminant dichromatic reflection model considers the effect of ambient light (i.e. a blue sky) in the pixel color value, not just the body color and surface reflection color.